

Ex.8 Producer consumer problem

Develop a C program to Implement producer - consumer problem. M producers and N consumers share a common buffer. Illustrate the output in multiple windows

AIM:

To Develop a C program to Implement producer - consumer problem. M producers and N consumers share a common buffer. Illustrate the output in multiple windows

CODE:

Main.c

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int c;
    printf("\n\t\t.....Producer Consumer.....\n");
    do {
        printf("\n1. Producer\n2. Consumer\n3. Exit\nEnter choice: ");
        scanf("%d", &c);
        if (c == 1) {
            system("gnome-terminal -- ./pd &");
        } else if (c == 2) {
            system("gnome-terminal -- ./cs &");
        } else if (c != 3) {
            printf("Invalid choice!\n");
        }
    } while (c != 3);
    return 0;
}
```

Producer.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/shm.h>
#include <sys/ipc.h>

#define BUFFER_SIZE 5

int main() {
    key_t k1 = ftok(".", 65);
```

```

key_t k2 = ftok(".", 66);
key_t k3 = ftok(".", 67);

int k1id = shmget(k1, sizeof(int), 0666 | IPC_CREAT);
int k2id = shmget(k2, BUFFER_SIZE * sizeof(int), 0666 | IPC_CREAT);
int k3id = shmget(k3, sizeof(int), 0666 | IPC_CREAT);

int *mutex = (int*)shmat(k1id, NULL, 0);
int *buffer = (int*)shmat(k2id, NULL, 0);
int *index = (int*)shmat(k3id, NULL, 0);

if (mutex == (void*)-1 || buffer == (void*)-1 || index == (void*)-1) {
    perror("shmat");
    exit(1);
}

*mutex = 0;
*index = 0;

int choice, value;

printf("\n--- PRODUCER ---\n");

do {
    if (*mutex == 0) {
        *mutex = 1;

        if (*index < BUFFER_SIZE) {
            printf("Enter value: ");
            scanf("%d", &value);

            buffer[*index] = value;
            (*index)++;

            printf("Produced. Buffer size: %d\n", *index);
        } else {
            printf("Buffer FULL!\n");
        }
    }

    *mutex = 0;
} else {
    printf("Consumer is working... Try later\n");
}

```

```

printf("Continue? (1/0): ");
scanf("%d", &choice);

} while (choice != 0);

shmdt(mutex);
shmdt(buffer);
shmdt(index);

return 0;
}

```

Consumer.c

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/shm.h>
#include <sys/ipc.h>

#define BUFFER_SIZE 5

int main() {
    key_t k1 = ftok(".", 65);
    key_t k2 = ftok(".", 66);
    key_t k3 = ftok(".", 67);

    int k1id = shmget(k1, sizeof(int), 0666 | IPC_CREAT);
    int k2id = shmget(k2, BUFFER_SIZE * sizeof(int), 0666 | IPC_CREAT);
    int k3id = shmget(k3, sizeof(int), 0666 | IPC_CREAT);

    int *mutex = (int*)shmat(k1id, NULL, 0);
    int *buffer = (int*)shmat(k2id, NULL, 0);
    int *index = (int*)shmat(k3id, NULL, 0);

    if (mutex == (void*)-1 || buffer == (void*)-1 || index == (void*)-1) {
        perror("shmat");
        exit(1);
    }

    int choice;

    printf("\n--- CONSUMER ---\n");

    do {

```

```

if (*mutex == 0) {
    *mutex = 1;

    if (*index > 0) {
        printf("Buffer before consume:\n");
        for (int i = 0; i < *index; i++) {
            printf("%d ", buffer[i]);
        }

        printf("\nConsumed: %d\n", buffer[*index - 1]);
        (*index)--;

        printf("Buffer after consume:\n");
        for (int i = 0; i < *index; i++) {
            printf("%d ", buffer[i]);
        }
        printf("\n");
    } else {
        printf("Buffer EMPTY!\n");
    }

    *mutex = 0;
} else {
    printf("Producer is working... Try later\n");
}

printf("Continue? (1/0): ");
scanf("%d", &choice);

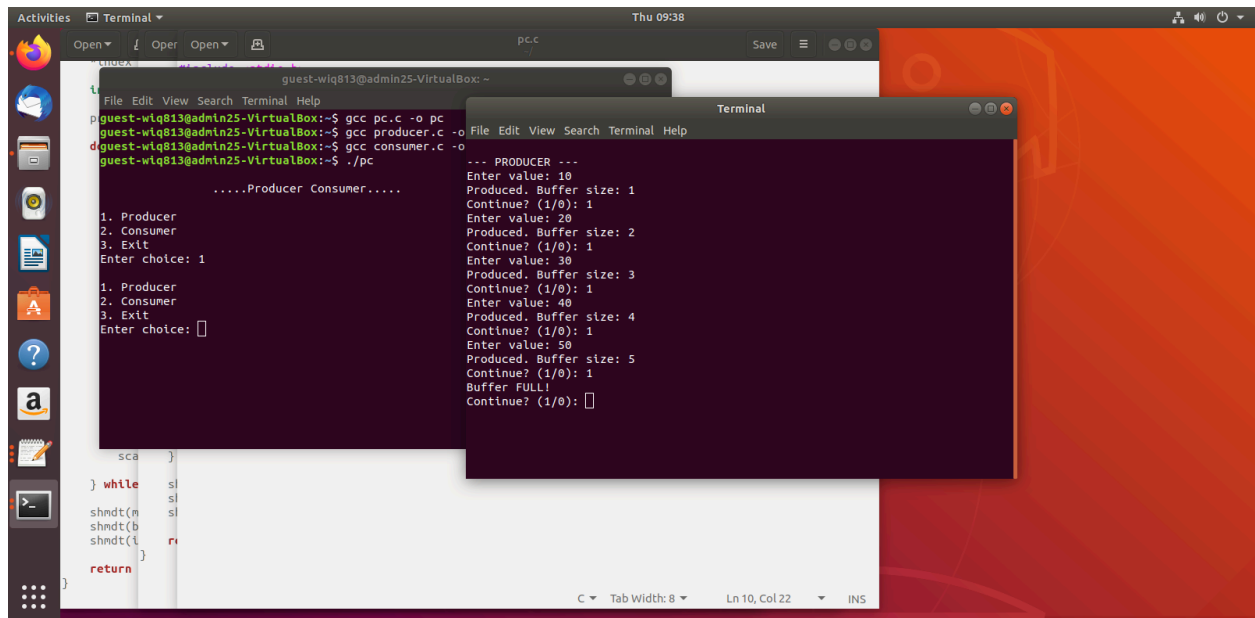
} while (choice != 0);

shmdt(mutex);
shmdt(buffer);
shmdt(index);

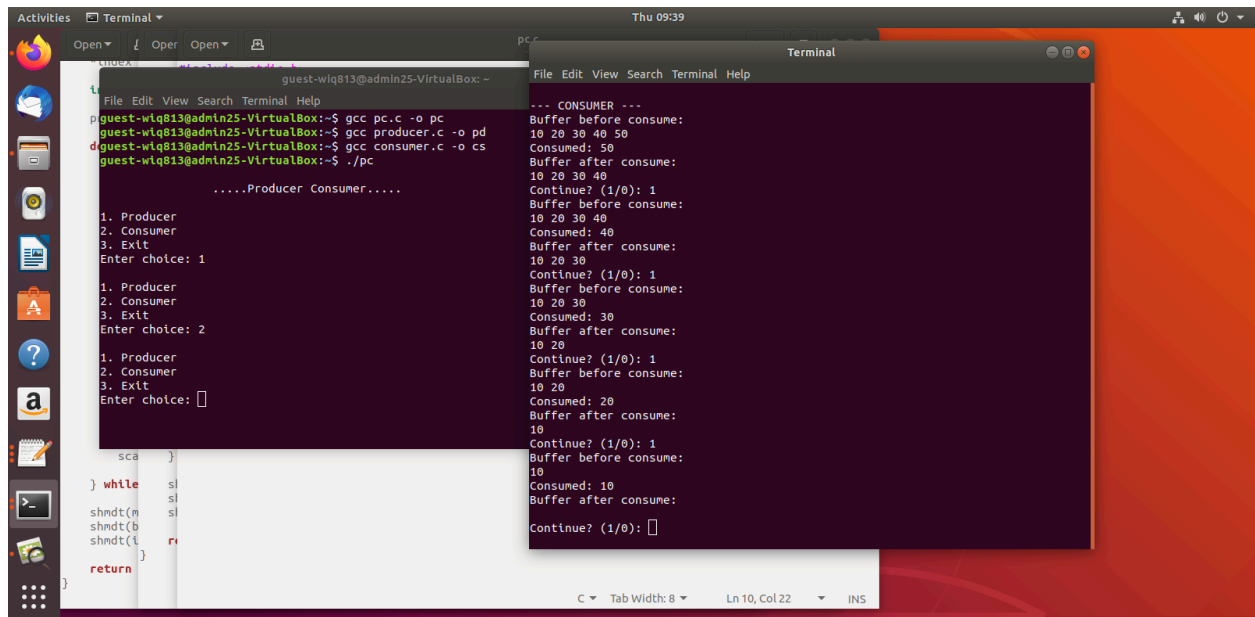
return 0;
}

```

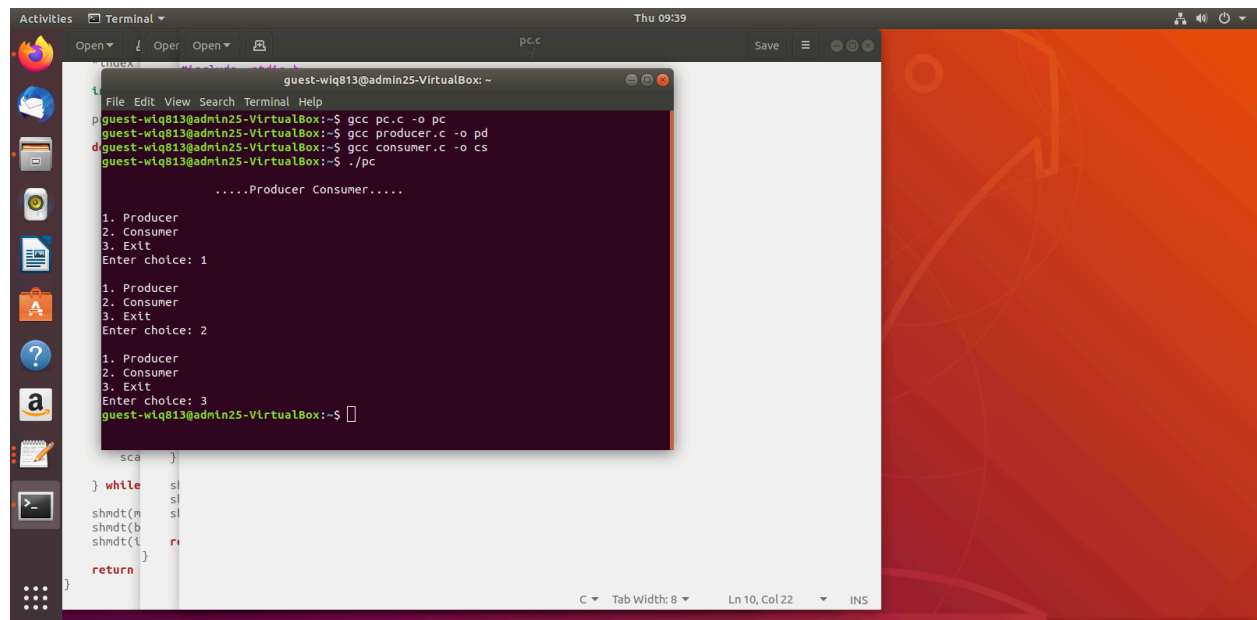
OUTPUT SCREENSHOT:



```
guest-wiq813@admin25-VirtualBox: ~  
File Edit View Search Terminal Help  
guest-wiq813@admin25-VirtualBox:~$ gcc pc.c -o pc  
guest-wiq813@admin25-VirtualBox:~$ gcc producer.c -o pd  
guest-wiq813@admin25-VirtualBox:~$ gcc consumer.c -o cs  
guest-wiq813@admin25-VirtualBox:~$ ./pc  
.....Producer Consumer.....  
1. Producer  
2. Consumer  
3. Exit  
Enter choice: 1  
1. Producer  
2. Consumer  
3. Exit  
Enter choice:   
Enter value: 10  
Produced. Buffer size: 1  
Continue? (1/0): 1  
Enter value: 20  
Produced. Buffer size: 2  
Continue? (1/0): 1  
Enter value: 30  
Produced. Buffer size: 3  
Continue? (1/0): 1  
Enter value: 40  
Produced. Buffer size: 4  
Continue? (1/0): 1  
Enter value: 50  
Produced. Buffer size: 5  
Continue? (1/0): 1  
Buffer FULL!  
Continue? (1/0):   
sca )  
} while si  
shndt(m si  
shndt(b si  
shndt(i ri  
return  
C Tab Width: 8 Ln 10, Col 22 INS
```



```
guest-wiq813@admin25-VirtualBox: ~  
File Edit View Search Terminal Help  
guest-wiq813@admin25-VirtualBox:~$ gcc pc.c -o pc  
guest-wiq813@admin25-VirtualBox:~$ gcc producer.c -o pd  
guest-wiq813@admin25-VirtualBox:~$ gcc consumer.c -o cs  
guest-wiq813@admin25-VirtualBox:~$ ./pc  
.....Producer Consumer.....  
1. Producer  
2. Consumer  
3. Exit  
Enter choice: 1  
1. Producer  
2. Consumer  
3. Exit  
Enter choice: 2  
1. Producer  
2. Consumer  
3. Exit  
Enter choice:   
sca )  
} while si  
shndt(m si  
shndt(b si  
shndt(i ri  
return  
C Tab Width: 8 Ln 10, Col 22 INS  
--- CONSUMER ---  
Buffer before consume:  
10 20 30 40 50  
Consumed: 50  
Buffer after consume:  
10 20 30 40  
Continue? (1/0): 1  
Buffer before consume:  
10 20 30 40  
Consumed: 40  
Buffer after consume:  
10 20 30  
Continue? (1/0): 1  
Buffer before consume:  
10 20 30  
Consumed: 30  
Buffer after consume:  
10 20  
Continue? (1/0): 1  
Buffer before consume:  
10 20  
Consumed: 20  
Buffer after consume:  
10  
Continue? (1/0): 1  
Buffer before consume:  
10  
Consumed: 10  
Buffer after consume:  
Continue? (1/0):   
C Tab Width: 8 Ln 10, Col 22 INS
```



```
guest-wiq813@admin25-VirtualBox: ~  
guest-wiq813@admin25-VirtualBox:~$ gcc pc.c -o pc  
guest-wiq813@admin25-VirtualBox:~$ gcc producer.c -o pd  
guest-wiq813@admin25-VirtualBox:~$ gcc consumer.c -o cs  
guest-wiq813@admin25-VirtualBox:~$ ./pc  
.....Producer Consumer.....  
1. Producer  
2. Consumer  
3. Exit  
Enter choice: 1  
1. Producer  
2. Consumer  
3. Exit  
Enter choice: 2  
1. Producer  
2. Consumer  
3. Exit  
Enter choice: 3  
guest-wiq813@admin25-VirtualBox:~$
```

RESULT:

Thus successfully implemented producer - consumer problem.